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Abbreviation

FMCG	Fast Moving Consumer Good
RM/PM/FG	Raw Material/Packing Material/Finished Good
SLEDB	Sri Lankan Export Development Board
PC	Personal Care
SWOT	Strengths Weaknesses Opportunities Threats
HR	Human Resource
HC	Head Count
IT	Information Technology
PP	Pallet Positions
SKU	Stock Keeping Unit
NPD	New Product Development
SOP	Standard Operating Procedure
PPE	Personal Protective Equipment
ERP	Enterprise Resource Plan
VAT	Value Added Tax
VSM	Value Stream Mapping
HTMS	Hemas Transportation Management System
GMP	Good Manufacturing Practice
GWP	Good Warehousing Practice

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CHAPTER 01- INTRODUCTION TO TRAINING ESTABLISHMENT

1.1 Introduction to Hemas Group of Companies

1.1.1 Origin of Hemas Group of Companies

Hemas is a leading diversified conglomerate in Sri Lanka, with a focus on four key sectors; FMCG, Healthcare, Transportation and Leisure. The Group was founded by Sheikh Hasannally Esufally (MBE) as a pharmaceuticals and trading enterprise. Over the ensuing decades the organization has transformed itself into one of Sri Lanka's largest and most respected firms. The Group has several recognizable brands including Baby Cheramy, Clogard, and Kumarika. Products and services of Hemas touch the lives of millions of people from new-born babies to large international businesses. Following part of the report describes vision, mission and core values of Hemas group of companies. Following diagram represents its successful history in a conceptual manner.

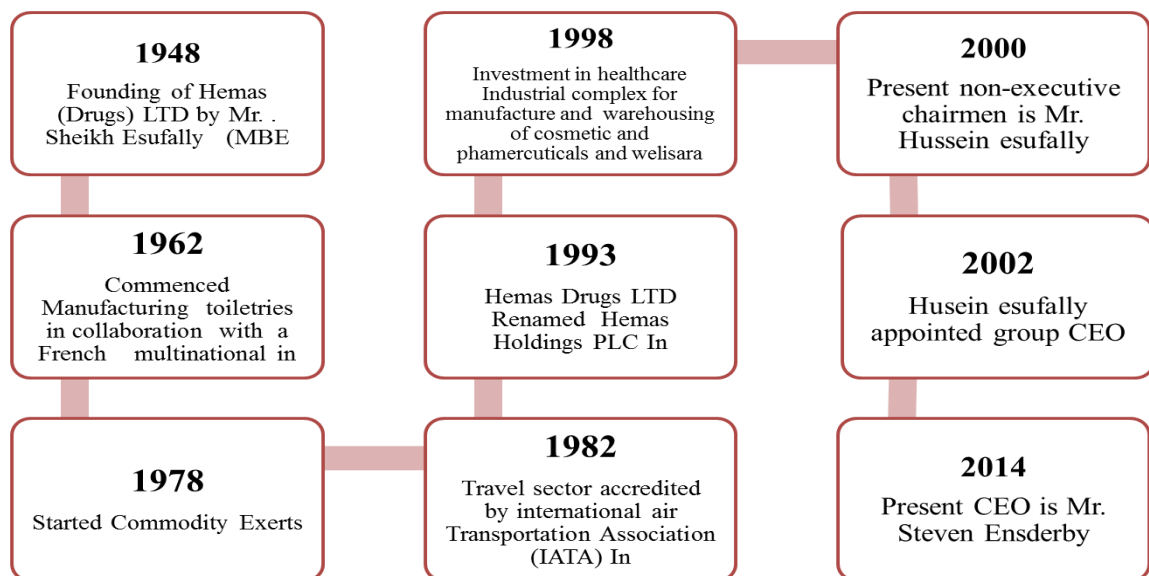


Figure 1.1: History of Hemas Group of Companies

1.1.2 Vision of Hemas Group of Companies

Their prime vision is to passionately deliver outstanding products and services thus enriching the lives of our customers and creating superior value to our shareholders. Their business model compromised with;

1. Being a National leader in Personal Care and Healthcare solutions
2. Investing in Growth industries with Potential for Superior Value Creation
3. Establishing a Regional footprint Through a Team of Passionate, Capable and Empowered People.

1.1.3 Mission of Hemas Group of Companies

Each individual in Hemas are committed to delivering to their shareholders the best ever returns a listed company has given an over the first seven years of listing for their employees. In addition, they are committed to creating exciting new opportunities and raising a share of wealth. For their business partners, they will deliver excellence in each and every time because they are a Responsible corporate citizen close to the lives of the community to the state and they are future winner with the nation and heart.

1.1.4 Core Values of Hemas Group of Companies

Hemas reveres its moral compass: its values as follows. It is their values that hold their together in meeting success, in providing their customers with quality products and services that they deserve.

1. Passion for Customer
2. Obsession for Performance
3. Driven by Innovation
4. Concern for People

1.2 Introduction to Hemas FMCG Sector

1.2.1 Business Overview

Hemas Group of Companies mainly operates in six sectors namely Fast Moving Consumer Goods (FMCG), HealthCare, Leisure, Transportation, Hemas Power and New Ventures. In addition, they provide working facilities nearly 5000 employees while contributing to the domestic economy as well. Out of those sectors, FMCG accounts as the largest sector at Hemas. Mainly because it is a leader in the personal care sector while Hemas brands encompass a range of product for babies and adult in hair care, skin care, toiletries, fragrances and oral care. Following figure provide snapshot of existing operations in FMCG sector as this is said to be the second most successful consumer brand in Sri Lanka. As it is mentioned, according to the 2018 statistics, Hemas FMCG sector accounts for a sale of 16.8 Billion rupees and 2.8 Billion rupees in international business segment. That shows how big Hemas is in 2018 and they have 3 flagship brands that accounts 5 million, 3 million and 2 million separately.



Figure 1.2: Business Snapshot of FMCG Sector

1.2.2 Market Overview

Currently Hemas FMCG sector is home for 15 Brands under the five categories of Personal Care, Home care, Personal Wash, Skin Care and Traded Brands.



Figure 1.3: Brands of Hemas FMCG Sector

1.2.3 International Business

Hemas International business has extended its horizons with serving to 12 countries around the world namely Maldives, Lebanon, Bangladesh, Belize, New Zealand, Zambia, Malaysia, Pakistan, Australia and the Middle East. In addition, Hemas brand names are well recognized across the continents and their distribution networks continues to grow. Hemas is involved in the following operations in international business:

1. Direct exports



Figure 1.4: 12 Direct Exporting Countries

2. Indirect exports

Hemas export their products through Sri Lankan Export Development Board (SLEDB) approved exporters. Following figure compromise the FMCG brands that are exported in the current exporting operations.



Figure 1.5: Exporting Brands

1.2.4 Current Status of Operation

Hemas currently has its own manufacturing facility in Dankotuwa and a distribution center in Welisara premise. One of the key driving factors of Hemas FMCG sector is to win the hearts and minds of consumers by delighting them with trusted high quality Personal Care products. Following part of the section describes the existing business scope and their respective operational achievements.

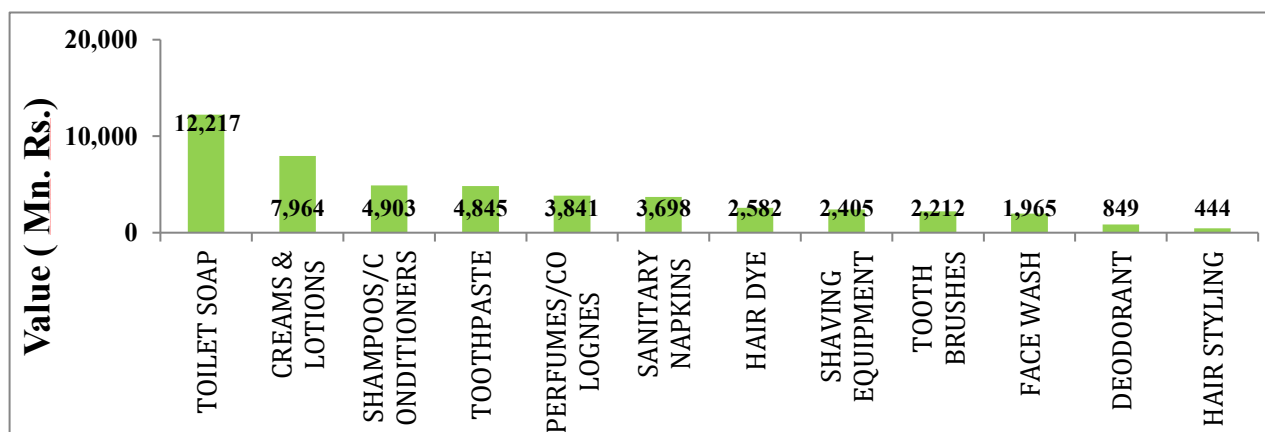


Figure 1.6: FMCG Market Composition

As it is shown in the above graph, Baby Cheramy and Velvet (Toilet soaps) segment hits the highest value in 2018 with 12,217 million rupees. Hair Dye (Godrej) accounts for the brand with highest volume growth with 21.9%. On the other hand, Velvet hand wash is having the lowest growth rate with -13.1%. Hemas Products reach to customers through more than 180,000 retail outlets and more than 350 Modern Trade outlets which are being fed by 1400 business partners around the country.

General Statistics about its brands

1. Baby Cheramy, 50+ year old brand, Leader in the Baby Care market at 45%+ share, present across sub categories of Baby Care – PW, Skin, Fragrance, Hair, Accessories
2. Clogard, 20+ year old brand, 28% share in Toothpaste, present across sub categories of Oral Care – Toothpaste, Toothbrush, Mouth Wash
3. Kumarika, 15+ year old brand, 53% share in Hair Oil, present across sub categories of Hair Care – Oil, Shampoo
4. Velvet, 10+ year old brand, 18% share in Adult PW, Leader in the Beauty Soap segment
5. Diva, 10+ year old brand, 24% share in Detergent Powder, present across sub categories of Laundry – Powders & Soap

1.2.5 Organizational Structure

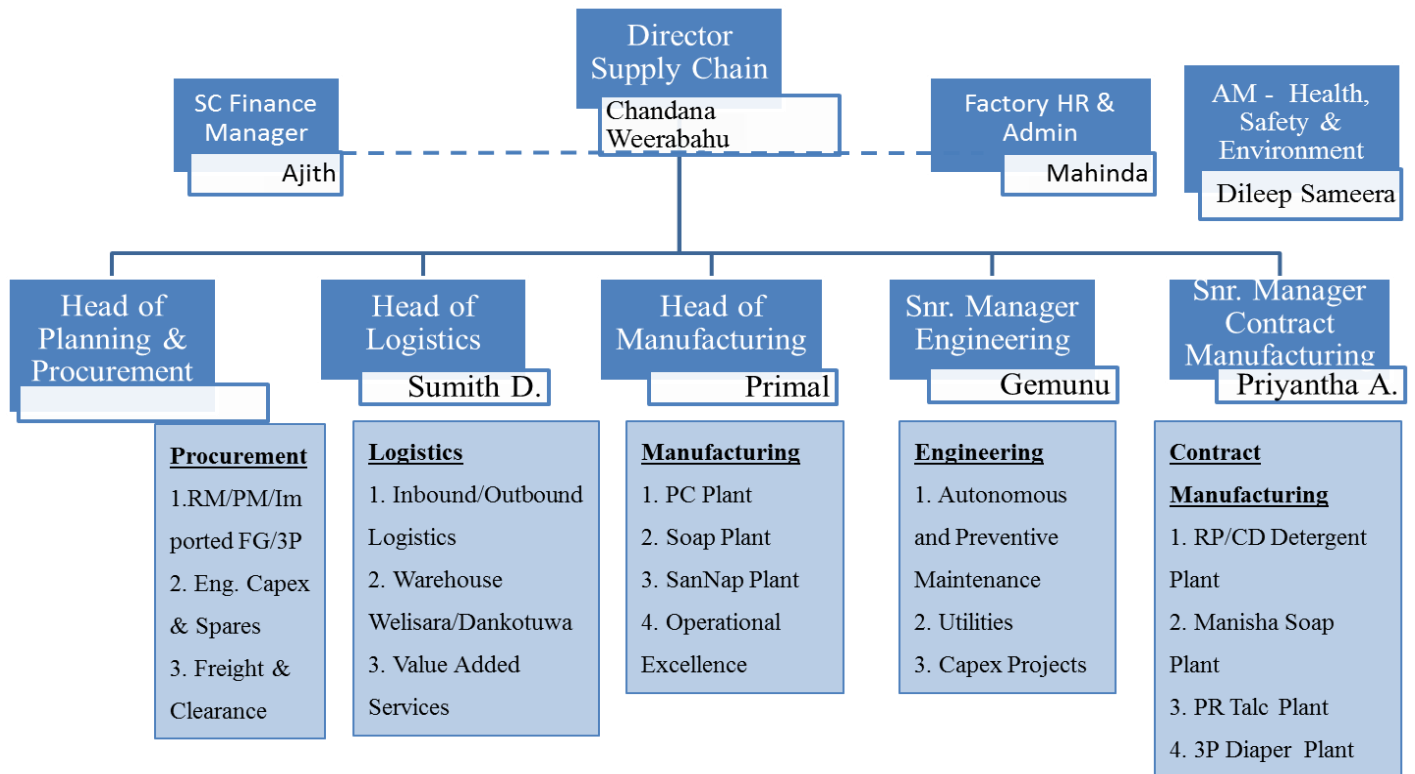


Figure 1.7: Organizational Structure of Hemas FMCG Sector

As it is shown the above diagram, each director has a span of control of 6 positions. They are namely Senior manager, Manager, Assistant manager, Senior executive, Executive and Junior executive. Therefore, Hemas has a very good management structure with consistent of levels in the hierarchy.

1.2.6 Usefulness of Organization to the Country

Hemas is mainly looking forward to understand the Sri Lankan consumer & lead change in the mind and behavior with faster, more relevant product offerings. They are mainly involve in green warehousing and manufacturing practices because their manufacturing plant is located near to a rural area (Dankotuwa). Therefore, Hemas is more concerning in reducing the carbon footprint as well. One of its prime objectives is to achieve a 30% market share by 2020. Following part of the report describes their key drivers and enablers for that 2020 vision.



Figure 1.8: Hemas Vision 2020

In my 6 months training period, I was able to work with following projects. Hemas has developed customized business models to serve different sectors of communities in the country. By doing so, they want to prepare a sustainable living commonplace and a healthy work force. Following projects are mainly targeted for the villagers because they are more experiencing a rural culture rather than an urbanized environment. Therefore, they are involving in reducing poverty and helping the economy to reach its horizons.

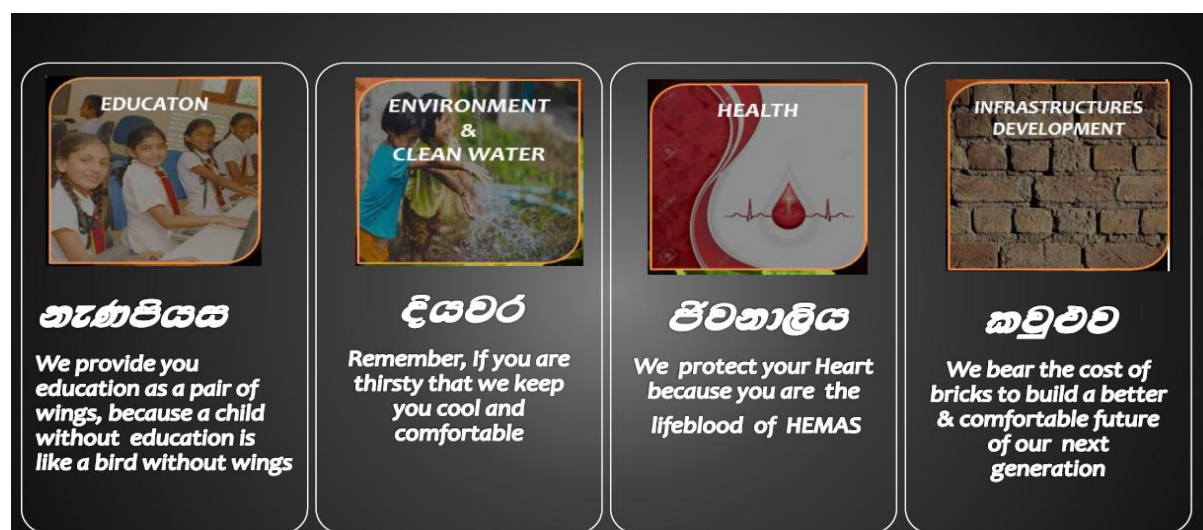


Figure 1.9: Way Forward CSR Plan

1.2.7 SWOT Analysis of Unilever Sri Lanka

Strengths	Weaknesses
1. Economic stability of the organization	1. Market slowdown and increased cost in getting topline
2. Recognition as a Group of Companies	2. Insufficient training on used software packages
3. Diversified product range of Hemas. (Baby Chera-my- 5million worth brand, Kumarika- 3million worth brand, Clogard, Velvert, Diva- 2million worth brands)	3. Operator's lack of knowledge
4. Competitive pricing and good quality due to global purchasing.	4. Centralized distribution center located in Welisara
5. Second most valuable company in the FMCG sector.	5. In house non-core operation (Labeling operation)
Opportunities	Threats
1. Manufacturing and Warehousing facilities can be automated	1. Lifestyle changes competing for share of wallet
2. Usage of modern and eco-friendly technologies.	2. Higher commodity prices, increased taxes, rupee devaluation
3. In house capabilities within the suppliers	3. Enhanced activities by Indian companies on the premise that ETCA will happen
4. Availability of technical inputs within the industry	4. Faster roll-out of premium products through Magnus (UL Global entry into SL)
5. Subsidies provided on manufacturing companies to enhance the trade balance (To focus more on exports)	5. Shopper seeking value in a high inflation environment.

Table 1.1: SWOT analysis for Hemas Organization

Hemas is said to be one of the leading organization in FMCG sector and they are extending their operations in International sector as well. In other words, they are trying to be a driving force to the trade balance that can give them more benefits in the long run. Since local competition stepping up their game by acquiring best talent, Hemas has to focus more on sustainability factors rather than developing as institute.

In addition, main competitor (Unilever) will engage and keep Hemas playing in the mud on selected categories and they can't just let those opportunities to fall upon. Centralizing non-core operations like labeling and looking for outsourcing options to focus more on core operations is really important to sustain in this highly competitive industry.

CHAPTER 02- TRAINING EXPERIENCE

2.1 Introduction to Training

I was selected as a Supply Chain Intern (in warehouse operations) at Hemas Dankotuwa manufacturing complex along with a batch of 9 other undergraduates from the Faculty of Engineering, University of Moratuwa. Training period was stated for 24 weeks starting from 25th June 2018 to 21st December 2018. During that period, I was mainly assigned to Warehouse section with one of our batch mates called Banusha. I hereby attach a map of Dankotuwa manufacturing complex to have a clear picture of our workings.

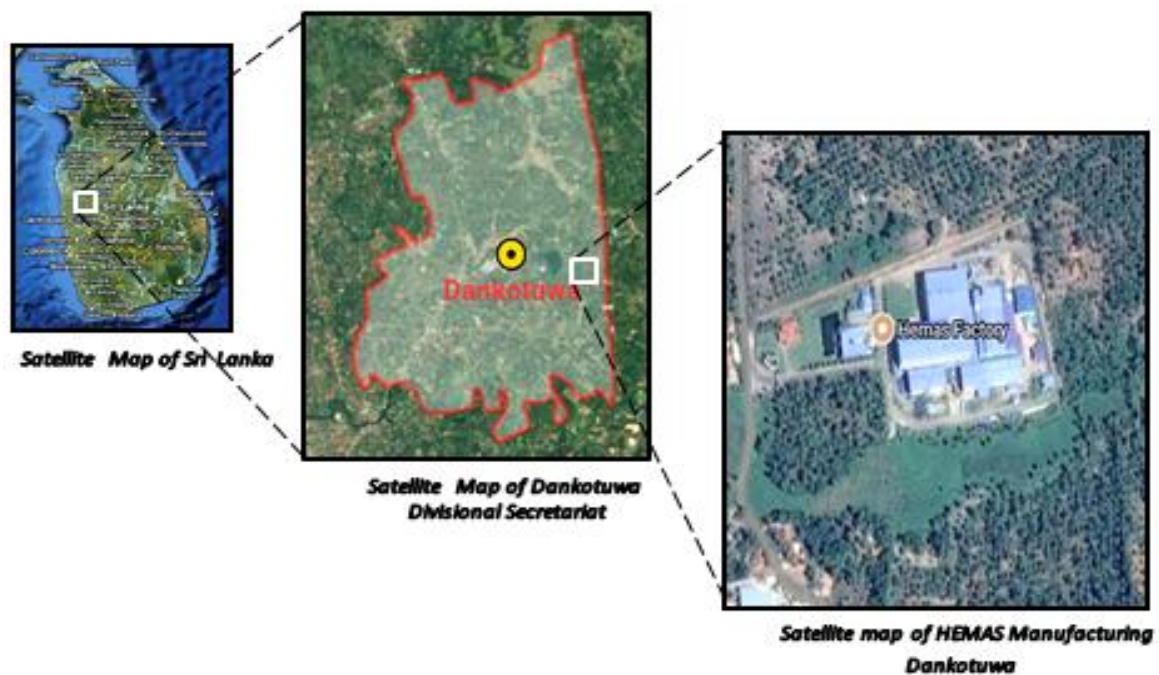


Figure 2.1: Satellite Map of Hemas Manufacturing Complex

During my first two weeks of the internship, I was introduced to factory layout, its health and safety procedures and department wise briefing. Those departments are namely HR, Contract Manufacturing, Planning, Procurement, IT, Account, Quality Assurance, Research & Development, Production Packing, Napkin, Mixing, Soap, Engineering, Stores (Warehouse). Other than those departments, Hemas manufacturing complex is consisted with 3 main production plants (Personal Care, Soap and Sanitary Napkins) and one R&D plant. In the Personal Care plant they manufacture Oral care, Skin care, Hair care, Baby care, Cologne & Perfumes whereas in Soap plant they do Baby and Beauty soap manufacturing.

As a Warehouse Intern I worked with the Hemas warehouse team which comprised of 5 sections namely Engineering stores, Raw Material stores, Packing Material stores, Labeling section and Shuttle service. I was able to work with each of the sections and I was assigned with different projects in each of the sections. The time period I was at Hemas was a crucial period for warehouse due to several reasons. During the month of September installation of a new racking system was implemented. As the warehouse intern, I had to carry lots of workings to smooth the transition process. In addition, I was asked to work with “Centralization of labeling process” project.

During my Internship I was involved in four major projects and with few minor projects. The four major projects are,

1. Statistics Inventory Model for RM & PM
2. Proposed Selective Racking System
3. Inventory Optimization through Safety Stock Level Reduction
4. Centralization of Labeling Process

Out of these four projects, I spent my most of the time with the centralization of labeling process. Because I had to conduct trials by myself while updating the top management. In addition, I did analyzing part and carried root-cause analysis to find the bottlenecks within the existing operation. Those workings really helped me to enhance my organizing, presentation and analytical skills.

2.2 Hemas Supply Chain

Hemas supply chain is divided into several operational aspects. They are namely: Supply planning, Procurement, Inbound logistics, Manufacturing, Engineering and Outbound logistics. Under the pillar of Supply planning, it's basically about the operational impacts regarding RM/PM/FG supply planning. It is operated both from the Dankotuwa and Colombo (Head office) premises. When it comes to procurement, they are having a strategic decentralized sourcing in order to reduce the supply risk.

When it comes to manufacturing, Hemas FMCG sector has a centralized manufacturing complex which is located in Dankotuwa area. Under that manufacturing complex, they have a Personal care plant, Soap plant, Sanitary Napkin plant. In addition to that, they are having few outsourced manufacturing plants including Diva manufacturing plant. I have included that information under the figure 2.3.

Supply Planning	Procurement	Inbound Logistics	Manufacturing	Engineering	Outbound Logistics
					
• FG • RM • PM • Imported FG • Production • 3P Conversion • Man power • Inventory management	• RM • PM • Imported FG • OS • Freight & Clearance • Promo items • Marketing expenditure	• Inbound Logistics • RM Warehouse • Dankotuwa	• PC Plant • Soap Plant • Sanitary Napkin Plant • Diva (3P) Plants • Operational Excellence	• Autonomous and Preventive Maintenance • Utilities • Capex Projects	• FG Warehouse • Welisara • Outbound Logistics • Value Added Services
HC - 5	HC - 15	HC - 38	HC - 316	HC - 34	HC - 109
Dankotuwa Colombo	Dankotuwa Colombo	Dankotuwa	Dankotuwa 3P Sites	Dankotuwa	Welisara

Figure 2.2: Supply Chain Operations in FMCG sector

Hemas is a fast growing company in the FMCG industry and figure 2.3 interprets their supply chain vision their respective action plans to achieve each of those strategic priorities. **“Reliable Supply of Assured Quality, Efficiently Delivered through Sustainable Operations with Great Systems & People”** is the main vision of FMCG sector of Hemas and they have 6 strategic priority pillars as well. They are namely Ensure availability, Assure quality, Optimize cost and cash, Build capacity and capability, Protect people & environment and Grow organizational potential. I have separately included actions for each of those priorities in the following diagram.

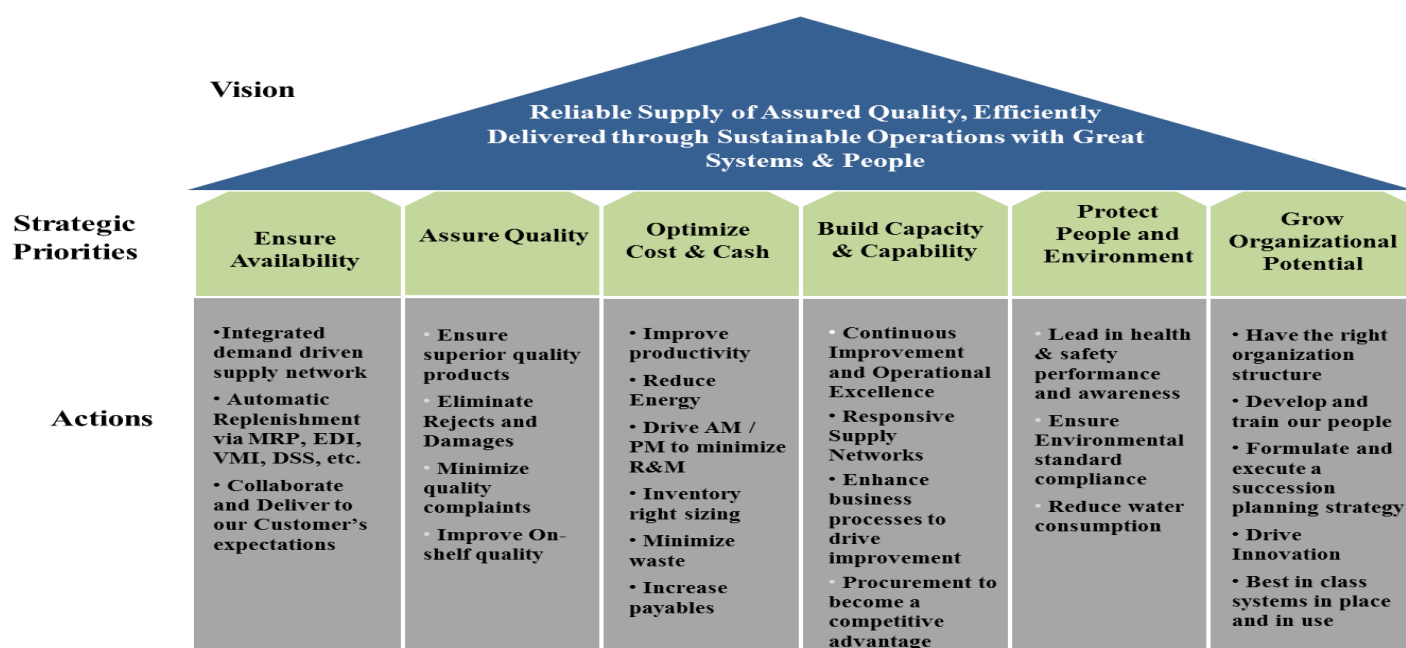


Figure 2.3: Supply Chain Vision with their respective Priorities

2.3 Hemas Manufacturing Complex

As it is described in the previous chapter, Hemas manufacturing complex in Dankotuwa is consisted with Human Resource, Contract Manufacturing, Planning, Procurement, IT, Accounts, Quality, Assurance, Research & Development, Production Packing, Napkin, Mixing, Soap, Engineering and Stores departments. Following part of the report describes each department's scope, sub sections and department Key Performance Indicators.

2.3.1 Logistics Department

56 employees are currently working in Dankotuwa logistics department and that include casual carder as well. Raw and Packing materials handling, Batch dispensing, Shuttle Operation, Scrap Management, Material Handling Equipment monitoring, Labeling Machine, Spare parts inventory, Roots cutting operation are key activities that are taken place in the logistics department. Following graph interprets the 5 main sections that come under the main department.



Figure 2.4: Sub Sections in Logistics Department

2.3.1.1 RM/PM Section

RM and PM sections are said to be the two main sections in logistics department as nearly half of the employees are involved into those two sections. Following table provide an informative output regarding RM/PM Section. Total value of this section accounts for 535 million rupees and out of that 406 million rupees worth for RM and 129 million rupees worth for PM.

	SKU	Issues PP Per month	No. of Process Orders per month	Receipts in Pallets per month
Raw Material	350	970	950	670
Packing Material	1200	1500	600	1100
Total		1470	1450	1770

Table 2.1: RM/PM Sectional Figures

2.3.1.2 Shuttle Service

As per the calculations, average trip length per month is accounted to 5540kms. Their shuttle operation mainly consisted with 12 twenty feet trucks and all those vehicles are said to be hired vehicles. Summarized graph of shuttle operation is presented in the figure 2.6. Huper, PR and RP said to be the 3rd party manufacturing plants and DC, JKL and Spectra locations are used for storing purpose. Detailed information is attached with the figure 2.7.

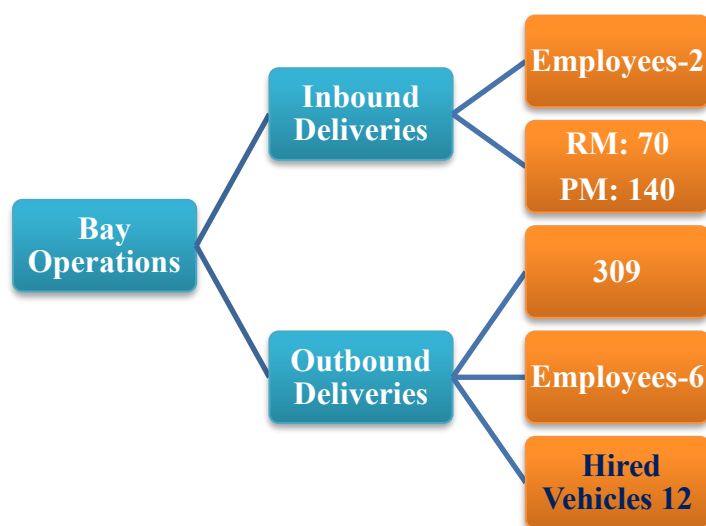


Figure 2.5: Summarized Shuttle Operation

Plant	Product/Material	Location
DC	FG	Welisara
Huper And pr	Diva	Waikala
PR	Nappy wash	Minuwangoda
RP	Diva	Bujjambola
3P(JKL)	RM,PM	Seeduwa
3P(Spectra)	RM,PM	Mabola

Figure 2.6: 3rd Party Service Locations

2.3.1.3 Labeling Section

Labeling operation is recognized as an non-core activity in the value adding process.

Labeling operation is happening in four locations. They are in Dankotuwa stores, Dankotuwa personal care plant, Welisara manual labeling and in 3rd party location. Therefore, Hemas is using both machine labeling and manual labeling options. On the other hand, 3 machines are available and 2 machines are located in Dankotuwa stores (Komark and Top print machines) and the remaining one is located in personal care plant (Premier machine). During the last two months of my internship, I was assigned to the project of “**Centralization of labeling process**”. There, I was mainly targeting the Premier machine because it has lots of bottlenecks and poor machine utilization. I have discussed those concerns more widely in the latter part of this report.

Machine	SKUs	Qty/ Month	Machine utilization	Location	Shift	People /Shift	Remarks
Komark	53	1,245,000	96%	Dankotuwa	8-8 24hrs	10	
TOP Print	3	374,152	54%	Dankotuwa	8-8 12 hrs.	6	KM 100ml Oil only
Premier	15	431,760	44%	Dankotuwa PC Plant	8-8 12 hrs.	9	
Manual	45	683,000		Welisara	7-7	28	
3p		178,000		Welisara			KM only
Totals	116	2,890,760					

Root inserts		364,000		Dankotuwa		8	KM 100ml Oil Only
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Table 2.2: Labeling Figures in September

2.3.2 Contract Manufacturing Department

Contract Manufacturing is the production of goods by one firm, under the label or brand of another firm. It provides such service to several (even competing) firms based on their own or the customers' designs, formulas, and/or specifications. This is also referred as private label manufacturing. Hemas is more concerned about contract manufacturing concept rather than out sourcing options.

There are 6 contractors manufacturing 43 SKUs at 7 geographical locations. 9 Employees from Hemas together with 222 employees from contractors are involved in this manufacturing process. Around 1800(60% of total company volume) metric tons a month is contributing with a value of Rs. 3.8 billion sales value per annum to the topline of the FMCG sector. On the other hand, Rs. 156 million worth of materials consumed a month at an operational cost of Rs. 16.2 million per month.

Hemas is using two contract manufacturing models to get the competitive advantage over other companies.

1. All materials and machineries are provided by Hemas at their book value. Plant is dedicated for Hemas and fully authorization is with Hemas. Open book costing is used to cost calculation. But the lower level employees are from Contract Manufacturing.
2. Only materials are provided by Hemas. Machines and sites are owned by the contractor with the full authorization. Employees are from Contract Manufacturing. In most cases actual cost isn't shared to achieve the win-win situation.

In addition, contract manufacturing is more likely to be a volatile business concept as following factors can be identified as challenges arising for an effective manufacturing. Those challenges are right-sizing the contract manufacturer selection, forecasting with reasonable accuracy, maintaining consistent quality, eliminating pricing surprises, defining an exit path with the contract manufacturer, controlling material losses, delivering the product on schedule, ensuring compliance with environmental directives.

2.3.3 Innovation Department

Hemas Innovation center is also referred to as Research and Development Center. Following activities can be recognized as key functions of this department. They are New Product Development s (NPDs), New Idea Generation, Specifications & Standards, Alternative Supplier / RM,PM developments, Cost Saving Projects, Knowledge Sharing, Trouble Shooting and Regulatory works.

Following figure provide a brief introduction about how R&D department create frameworks to check their successfulness.

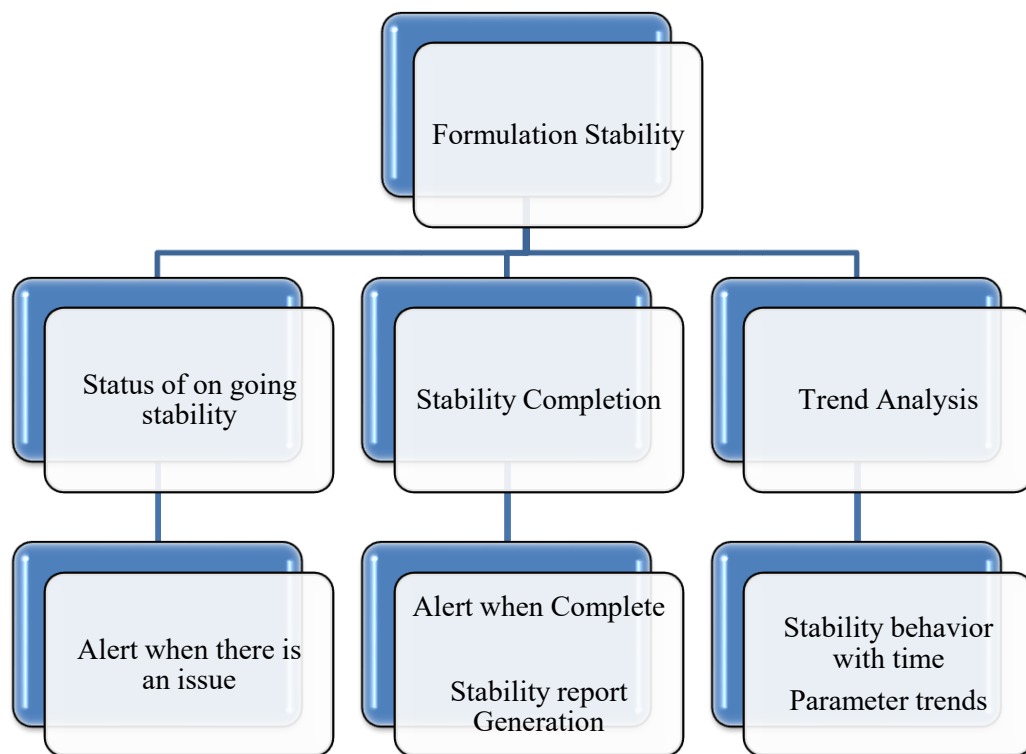


Figure 2.7: Conceptual Framework to Check Formulation Stability

2.4 Warehouse Operation

The warehousing operation of Hemas is partially out sourced. That is because they are having few pallet positions in third party warehouse locations. Following diagram represents the flow and key functions in Hemas warehouse operation.

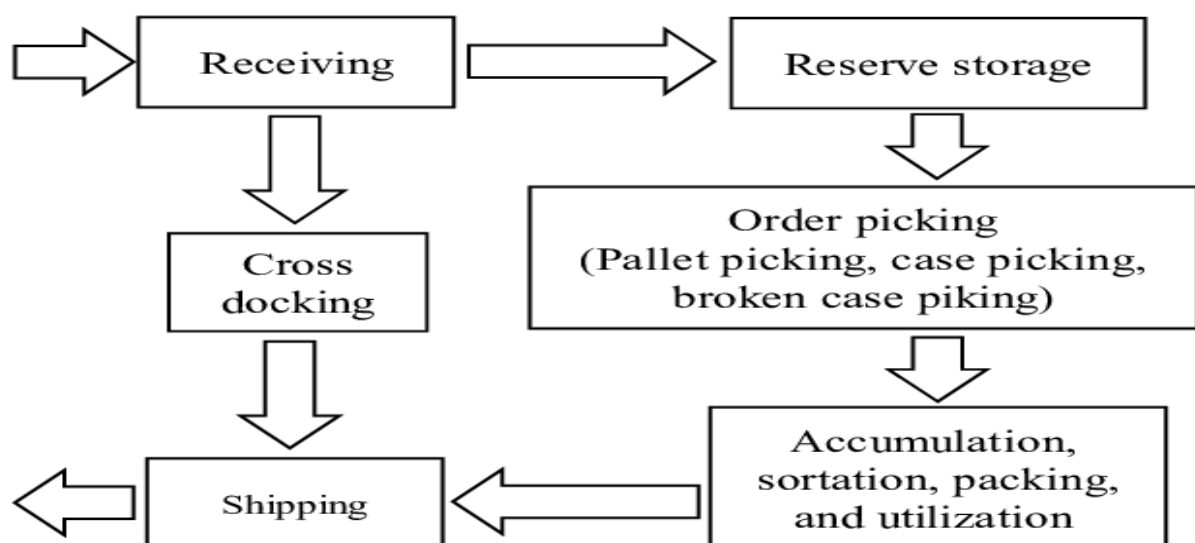


Figure 2.8: Main Warehouse Functions

2.4.1 Replenishment

Process Step	Procedures / Documents	Responsibility
Receiving of consignment	Email	Logistics executive/ Logistics officer
Available Not available Availability of relevant documents	Invoice or Delivery note or Advice of Dispatch or Packinglist	Material receiver
Return Vehicle without unloading	Invoice or Delivery note or Advice of Dispatch	Logistics executive/ Logistics officer
Unload consignment	Invoice or Delivery note or Advice of Dispatch	Unloading crew
No Yes Discrepancy	Invoice or Delivery note or Advice of Dispatch	Material receiver
Excess Shortage Excess or Shortage		Material receiver
Inform the commercial dept and request to send the short qty while accepting available qty		Logistics executive/ Logistics officer
Return the excess qty.	Gate pass/ Invoice or Delivery note or Advice of Dispatch	Material receiver
No Yes Damages		Material receiver
Separate the damages and Inform to commercial/ QA Dept.	Non Conformity Report (NCR)	Unloading crew/ Material receiver
Act based on commercial advice	Invoice or Delivery note or Advice of Dispatch	Material receiver
Record the right qty. in the receiving register	Receiving register	Material receiver
Paste quarantine labels		Material receiver/ unloading crew
Store in quarantine area and raise system GRN	Good Receiving Note (GRN)	SAP operator
Taking Samples and send to laboratory		Sample taker/ Lab assistant
Yes No QA approval		QA executive
Paste hold/rejected labels and block in the system	Non Conformity Report (NCR)	QA officer
Move into rejected area		Material handling team
Paste accepted labels and release from the system		QA officer
Store for unrestricted use		Material handling team

Figure 2.9: Material Receiving and Storing Process

During my internship period, I was introduced to their replenishment operation and I was asked to prepare Standard Operating Procedure (SOPs) for their main operations. That is how I derived those flow diagrams with the help of warehouse team. One of the main norms included in their replenishment is that there should be stock for 2 weeks requirement for Packing materials at any given time and stock for 1 month requirement for Raw materials at any given time. Safety stock policy is also aligned with this norm and procurement department do a significant job to cater these requirements.

2.4.2 Issuing Process

Issuing process is linked to batch weighing process because RM section issue materials to Mixing department. Within the given time period, I was able to find following opportunities and threats in the issuing process.

Opportunities- SAP system, Appropriate PPEs, Material handling equipment, Sophisticated weighing machines, Competent staff and employees.

Threats- Accidents while driving and walking, Space availability within the premise.

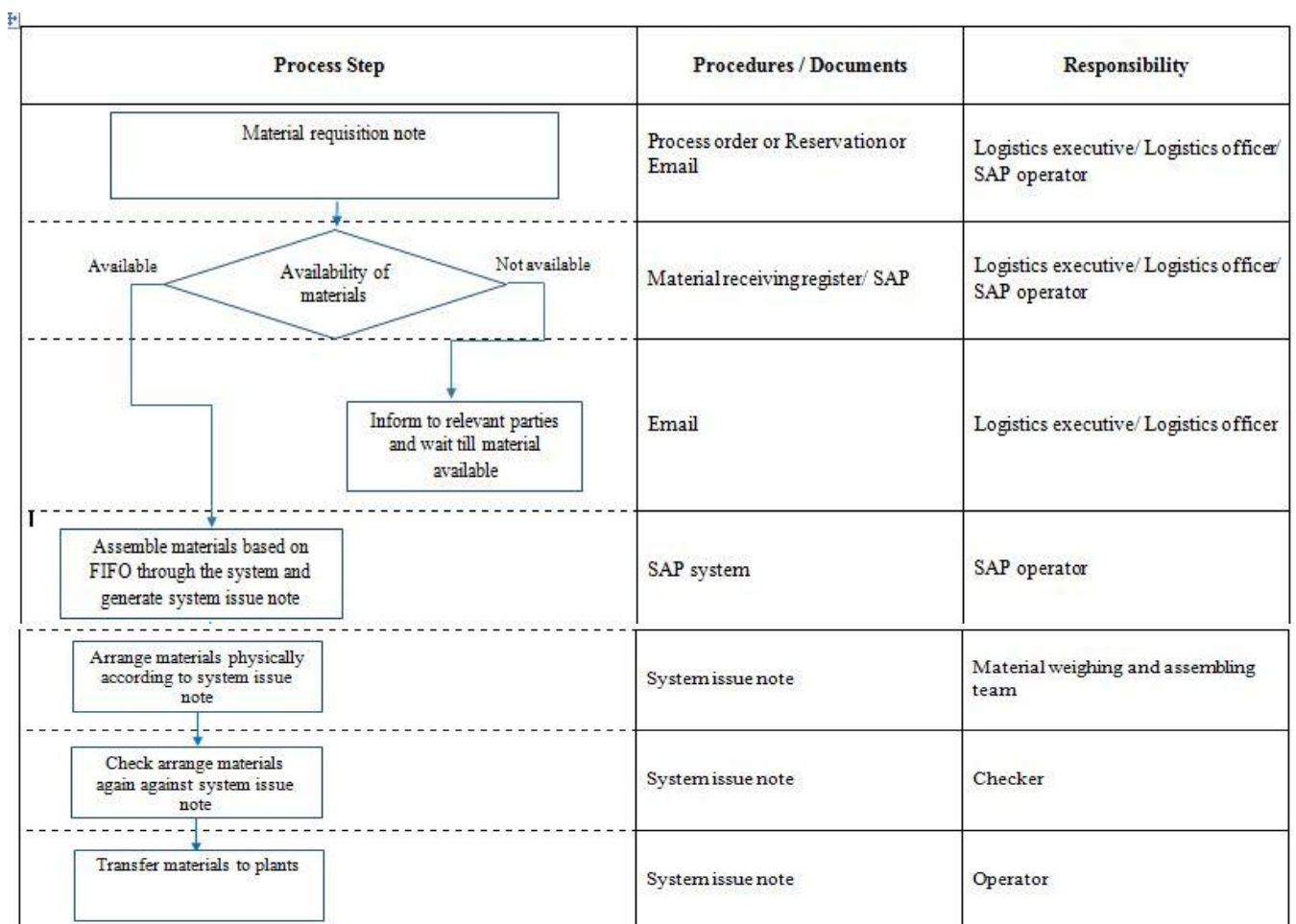


Figure 2.10: Material Issuing Process

2.4.3 Scrap Sales Process

Scrap refers to waste that either has no economic value or only the value of its basic material content recoverable through recycling. Manufacturers do not intend to produce scrap.

However due to the nature of the production process scrap will be generated and management of scrap should be done to minimize the negative impact on company's profit.

Depending on the type of scrap it may be;

- Recycled back in to the production process
- Sold as a revenue generating product
- Given to a third party at a fee, to recycle/dispose

Development and implementation of a procedure for scrap sales is done with the following objectives.

1. Ensure transparency at all stages- Selecting buyers, sale of scrap and recognition in books of account.
2. Ensure that effective and efficient internal controls are being followed.
3. Flexibility and simplicity of the procedure.
4. Integrating scrap sales procedure in to SAP ERP.
5. Adherence with reporting requirements.

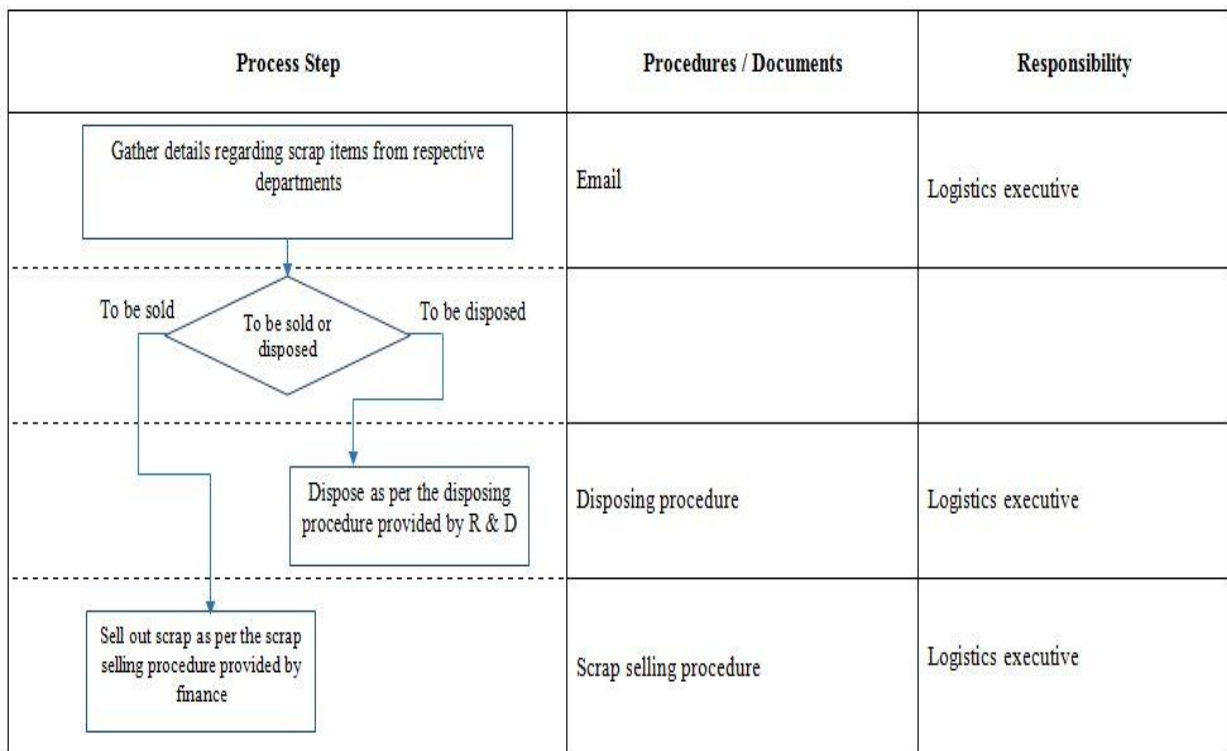


Figure 2.11: Scrap Sales Procedure

Document	Stage	Prepared by	Authorized by	
Tender approval document	Obtaining higher level approval for the selected Quotations.	Scrap Sales Committee	Supply Chain Director	
List of scrap with economic value	Identification of scrap with economic value	Officer responsible	Head of Manufacturing	
Invoice	Point of sale	Officer responsible	Head of Logistics / Head of Manufacturing	Supply Chain Finance Manager
Gate pass	Handing over the consignment to the buyer for dispatch	Officer responsible	Designations specified by Human Resource Department for Gate pass authorization.	
Gate Pass raised earlier	Authorizing the buyer to take the goods from factory premises		Security Officer who checked the consignment.	
Private placement approval document	Obtaining higher level approval for the selected Buyers of a private placement	Scrap Sales Committee	Supply Chain Director	
Credit approval document	Obtaining higher level approval for granting credits to a Buyer of scrap items	Scrap Sales Committee	Supply Chain Director Finance Director	
Approval to sell a rejected raw material	Obtaining approval to sell a raw material that has been rejected due to a quality failure caused by the Supplier		Supply Chain Director Finance Director	

Table 2.3: Documents Raised at Each Level

Other than these documents raised in each level, Hemas is trying to follow a “**green policy**” for Scrap selling process. I was introduced to that process and following points describe key findings with my own workings.

Invoice to be raised in SAP – Once the buyers confirm the purchase, responsible Officer need to carry out the following steps;

1. Sales orders should be raised in SAP. If the scrap item is not included in inventory following steps should be followed before raising a sales order.

a. Create a dummy material and material code. Example- SISW- Sweep waste of Soap plant.

b. Assign the item category as TAX. Separate location can be created for scrap items under a unique code.

2. At the time of raising the invoice, responsible Officer should raise it in SAP. Once the invoice is generated, accounting document will be generated where the value will be debited to cash/customer account and sundry income, VAT and NBT will be credited.

3. Manual Gate pass should be raised as per the instructions in step 5 of; General guidelines for the collection and sale of scrap.

4. Responsible Officer should raise a PGI in SAP. Items which were in inventory originally will be deducted from the storage location, “blocked” and the dummy materials will be deducted from the storage location, “scrap”.

5. Senior Warehouse Executive should send scanned copies of the bank deposit slips to Senior Executive Finance, mentioning the relevant invoice number.

6. Senior Executive -Finance should match the bank deposit slips with the system invoices and send the relevant information to Shared Service Unit.

7. Shared Service Unit will post the journal entries.

In addition to that, those findings will have the following disadvantages.

- Complexity – Dummy materials need to be created and removed by raising a PGI.
- Burden of additional work

Therefore, I have suggested that above complexities can be avoided if an invoice can be raised without matching it with an inventory item in the system. Then there won't be a need of raising sales orders and PGIs.

2.5 Project 1 – Statistics Inventory Model for RM & PM

During my first few weeks of internship, I was introduced to their delivery schedule, purchasing policies and to their inventory holding policies. I was able to get the help warehouse management team and I was able to arrive at a re-order point and maximum stock level for each SKUs. There, I considered delivery schedule to determine the maximum pallet requirement.

2.5.1 Methodology used in the Project

1. During this period I was referring to lots of academic resources and finally was able to find a suitable way to calculate maximum stock level for each SKU. Expectation of Lead time demand and Variance of Lead time demand was separately calculated in order to arrive at the Re-Order point (ROP) level.

$Safety\ stock = k(MAD) \times \sigma$ k = a multiplier for standard deviation of demand from desired service level σ = The demands standard deviation during lead time. It shows the variation of demand during the time it takes to refill stock. $\sigma = \sigma_t \times \sqrt{LT}$ σ_t = Standard deviation of demand during time LT = Lead time, time until the stock is refilled	$E[D_L] = E\left[\sum_{i=1}^L D_i\right] = \mu_L \mu_D$ $V[D_L] = V\left[\sum_{i=1}^L D_i\right] = \mu_L \sigma_D^2 + \sigma_L^2 \mu_D^2$
--	---

Figure 2.12: Equations Used in Statistics Model

Assumptions made to derive those equations

1. Lead time is constant (Variation of lead time is zero)
2. Lead time demand is randomly distributed
3. Storage is planned for maximum stock situations
4. Confidence level of 95%

2. After the calculation of ROP levels, I derived following equations to calculate maximum stock level at any given time for each SKU.

Order Quantity = Time between Orders * Average Monthly Consumption

Maximum Stock Level = Reserve level + Order Quantity

Quantity to be ordered = Order quantity + Minimum Order Level – Stock Balance

2.5.2 Outcome of the Project

1. By referring to above equations, I derived following platform and following part of the report describes key outcomes of this project.

desired ROP:

$$ROP = \mathbb{E}[D_L] + z_\alpha \sqrt{\mathbb{V}[D_L]} = \mu_D \mu_L + z_\alpha \sqrt{\mu_L \sigma_D^2 + \mu_D^2 \sigma_L^2}$$

Material Code	Description	Lead time	Future Avg.Usage	Variance (Demand)	Expectation Lead Time Demand	Variance Lead Time Demand	ROP
203560	LAMI TUBE CL TP 70g - NEW (2015)	60	338394	2535331774	676787	116782	793570
205079	CARTON BC SOAP FLORAL 100G - YR18	30	579856	15353207801	579856	203209	783065
203560	LAMI TUBE CL TP 70g - NEW (2015)	60	353795	2535331774	707590	116782	824372
203566	CARTON CL TP 70g - NEW (2015)	30	353795	2535331774	353795	82577	436373
204812	DV SOAP WRAPPER YELLOW 120G - YR17	30	940281	84703488750	940281	477303	1417584
204612	WRAPPER - VV SOAP SANDALWOOD 100g - YR17	15	448324	10021282711	224162	116089	340251
204608	WRAPPER - VV SOAP MILK 100g - YR17	15	838551	1.12282E+11	419275	388584	807859

Figure 2.13: Sample of the Excel File -1

2. Since I got only delivery schedule for RM, I only calculated maximum pallet requirement for RM. Following figure describes the first 9 maximum pallet required raw materials.

Material Code	Description	Future Monthly Consumption	Buffer stock	Order Quantity	Delivery Days	Pallet requirement
101054	CALCIUM CARBONATE (GROUND)	64,260	40662	32130	15	73
101293	FLUFF UNTREATED - GOLDEN ISLES G4881	48,423	48157	24211	15	56
100098	SORBITOL	63,310	36986	21103	10	55
100036	CAUSTIC SODA	82,464	42109	8246	3	53
101466	3D BUBBLE NON-WOVEN TOP SHEET 38GSM-90MM	3,135	2873.	3135	30	41
100108	TALCUM POWDER	32,594	30613	7605	7	41
100094	SILICA 165 THICKENING	8,526	4436	9240	30	35
100323	BC COLOGNE BULK - IMPORTED	20,249	4902	20248	30	32
100460	GALAXY LES 70	16,980	10533	8489	15	22

Figure 2.14: Sample of the Excel File -2

Detailed analysis has been attached in the appendix for the further reference.

2.6 Project 2 – Proposed Selective Racking System

2.6.1 Background of the project

Initial phase of this project (Proposed Selective Racking System) was launched in 2012 by implementing a racking system in RM/PM stores in Dankotuwa. I was recruited mainly for this project to cater the transmission of goods in a smooth way. Before this installation, Warehouse had 2400 net number of pallet positions and they were incurring a high cost at third party warehouses because those 2400 pallet positions were not sufficient to cater the high demand.

Outcome of the Installation process

1. Net increased pallet positions = 475
2. Total pallet positions= $1657+848+296+75 = 2875$

As it is mentioned, this installation provided 848 new pallet positions but it was a net increase of 475 pallet positions. Following figure is the map of installed racking system. Existing racking system is also included in the following map.

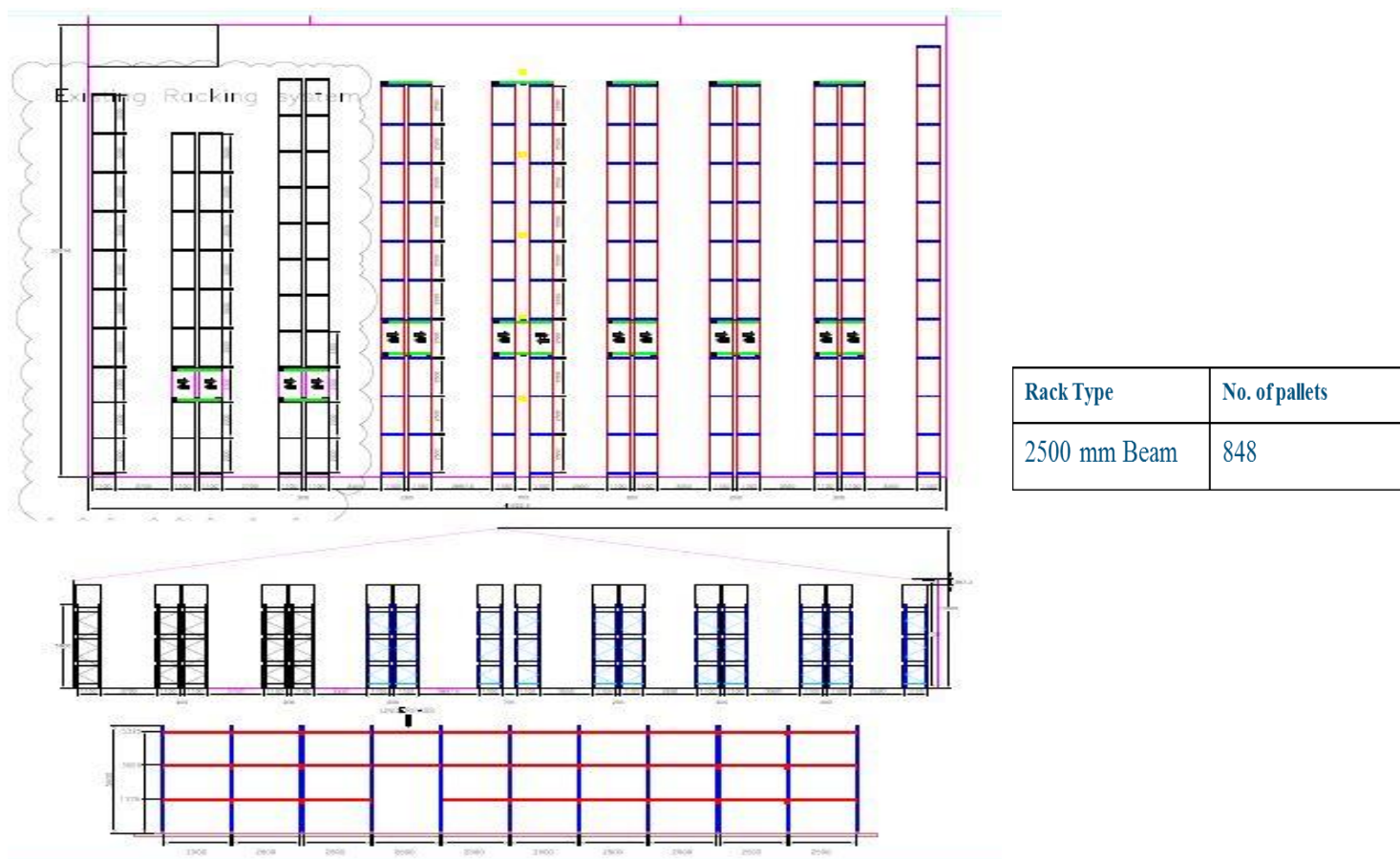


Figure 2.15: Hemas Racking Plan

2.6.2 Project Delivery

Main purpose of this project was to allocate specific space for each SKU according to their volume of usage and frequency of usage. Efficient and effective operational practices in FMCG industry are essential to achieve continuous improvement. That is where I have adapted advanced improvement opportunities to eliminate those disrupted practices.

2.6.2.1 ABC Inventory Categorization

ABC inventory categorization is a systematic way to recognize the importance of each SKU. Therefore, I conducted ABC categorization for Raw materials and Packing materials separately.

Raw Material ABC Categorization

With the help of warehouse team, I've conducted a high level analysis to determine the movement behaviors of Raw material SKUs at Hemas. Unit Issues, Service Unit Issues and Direct Issues data were considered in this analysis. As it is included in the graph, I conducted an SKU category wise analysis for 288 SKUs on ABC categorization based on the volumes of transactions from September 2017 to August 2018 and obtained the results in the tables.

ABC Categorization - Number of SKUs (% of Total SKUs)					Pallet Positions
	A	B	C	Total	
Drums	11(7.7%)	9(6.3%)	123(86%)	143	296
Bags	3(7.1%)	5(11.9%)	34(81%)	42	288
Reel	4(18.1%)	11(0.5%)	7(31.9%)	22	211
Bottles	1(5.0%)	4(20.0%)	15(75.0%)	20	29
Boxes	1(7.7%)	4(30.7%)	8(61.6%)	13	28
Buckets	2(22.2%)	1(11.1%)	6(66.7%)	9	11
Cans	1(2.9%)	3(8.8%)	30(88.3%)	34	53
Other	1(20%)	1(20%)	3(60%)	5	5
Total	24	38	226	288	921

ABC Categorization - % of Volume			
	A	B	C
Drums	80.9%	8.8%	10.3%
Bags	82.5%	10.5%	7%
Reel	78.5%	20.6%	0.9%
Bottles	76.7%	18.6%	4.7%
Boxes	83.4%	13.3%	3.3%
Buckets	82.4%	11.1%	6.5%
Cans	77.8%	9.6%	12.6%
Other	93.1%	5.9%	1.0%

Table 2.4: ABC Analysis for RM

The first table shows the number of SKUs and the percentage of total SKUs which belongs to each category under the stratification of ABC. The 'A' category represents the SKUs which accounted for high volume of usage, while the 'B' category and the 'C' category represent the SKUs which accounted respectively for medium volume usage and low volume usage. The values depicted in the second table shows the percentages of the volumes for each category under ABC stratification. For this analysis, only the items issued from RM Stores were considered and the returns to the RM Stores were not considered.

Packing Material ABC Categorization

As it is included in the graph, I conducted an SKU category wise analysis for 458 SKUs on ABC categorization based on the volumes of transactions from September 2017 to August 2018 and obtained the results in the tables. When doing this categorization, I divided each SKU according to their type namely Corr.Boxes, Bags, Caps, cartoons, Dividers, Glass Bottles, Inner Liners, Inserts, Neck tags, Pouch, Sachets, Tubes and Wrappers. I followed this procedure for Raw materials as well.

ABC Categorization - Number of SKUs (% of Total SKUs)				
	A	B	C	Total
Corr. Boxes	27(29%)	10 (10.8%)	56 (60.2%)	93
Bags	3 (25%)	3 (25%)	6 (50%)	12
Caps	18 (24.3%)	11 (14.8%)	45 (60.9%)	74
Cartoons	10 (5.0%)	13 (20.0%)	49 (75.0%)	72
Containers	20 (37.7%)	12 (22.7%)	21 (39.6%)	53
Dividers	5 (21.7%)	3 (13.1%)	15 (65.2%)	23
G/Bottles	2 (12.6%)	3 (18.7%)	11 (68.7%)	16
I/Lining	1 (20%)	2 (40%)	2 (40%)	5
Inserts	2 (28.6%)	2 (28.6%)	3 (42.8%)	7
N/Tags	8 (33.4%)	4 (16.6%)	12 (50%)	24
Pouch	6 (25%)	3 (12.5%)	15 (62.5%)	24
Sachets	7 (35%)	3 (15%)	10 (50%)	20
Tubes	4 (17.8%)	6 (27.2%)	12 (55%)	22
Wrappers	5 (38.5%)	3 (23%)	5 (38.5%)	13
Total	118	78	262	458

ABC Categorization - % of Volume			
	A	B	C
Corr. Boxes	80.1%	8.1%	11.8%
Bags	78.2%	17%	4.8%
Caps	80.8%	10.4%	8.8%
Cartoons	82.6%	11.9%	5.5%
Containers	79.2%	13.3%	7.5%
Dividers	84.2%	6.8%	9.0%
G/Bottles	77.5%	14.1%	8.4%
I/Lining	62.0%	35.3%	2.7%
Inserts	82.7%	11.7%	5.6%
N/Tags	80.0%	11.1%	8.9%
Pouch	80.0%	11.7%	8.3%
Sachets	77.0%	13.0%	10.0%
Tubes	78.0%	15.3%	6.7%
Wrappers	74.0%	16.5%	9.5%

Table 2.5: ABC Analysis for PM

The first table shows the number of SKUs and the percentage of total SKUs which belongs to each category under the stratification of ABC. The values depicted in the second table shows the percentages of the volumes for each category under ABC stratification. For this analysis, only the items issued from PM Stores were considered and the returns to the PM Stores were not considered.

2.6.2.2 FMS Inventory Categorization

According to FMS analysis done for RM and PM, I was able to determine Slow-moving items and Non-moving items. Following part of the report contains a comprehensive analysis of FMS inventory categorization. FMS categorization is based on the frequency of unit transactions from September 2017 to August 2018 and obtained the results in the below tables.

FMS Analysis - Number of SKUs (% of Total SKUs)				
	F	M	S	Total
Drums	11(7.7%)	53 (37%)	79 (55.3%)	143
Bags	13 (30.1%)	20 (47.6%)	9 (22.3%)	42
Reel	5 (22.7%)	10 (45.4%)	7 (31.9%)	22
Bottles	2 (10.0%)	3 (15.0%)	15 (75.0%)	20
Boxes	4 (30.7%)	4 (30.7%)	5 (38.6%)	13
Buckets	3 (33.3%)	4 (44.4%)	2 (22.3%)	9
Cans	4 (11.7%)	14 (41.1%)	16 (47.2%)	34
Other	1 (20.0%)	2 (40.0%)	2 (40.0%)	5
Total	43	110	135	288

Raw Material FMS Categorization

FMS Analysis – Frequency			
	F	M	S
Drums	> 1000	999 – 101	< 100
Bags	> 940	939 – 103	< 102
Reel	> 76	75 – 23	< 22
Bottles	> 763	762 – 103	< 102
Boxes	> 775	774 – 163	< 162
Buckets	> 203	202 – 108	< 107
Cans	> 514	513 – 102	< 101
Other	> 16	15 – 8	< 7

FMS Analysis - Number of SKUs (% of Total SKUs)				
	F	M	S	Total
Corr. Boxes	7(7.7%)	17 (37%)	69 (55.3%)	93
Bags	1 (8.4%)	6 (50%)	5 (41.6%)	12
Caps	6 (8.2%)	24 (32.4%)	44 (59.4%)	74
Cartoons	3 (4.2%)	8 (11.1%)	61 (84.7%)	72
Containers	8 (15%)	6 (11.4%)	39 (73.6%)	53
Dividers	4 (17.3%)	2 (8.8%)	17 (73.9%)	23
G/Bottles	1 (6.2%)	2 (12.5%)	13 (81.3%)	16
I/Lining	1 (20%)	1 (20%)	3 (60%)	5
Inserts	1 (14.2%)	2 (28.6%)	4 (57.2%)	7
N/tags	1 (4.1%)	11 (45.9%)	12 (50.0%)	24
Pouch	4 (16.7%)	5 (20.8%)	15 (62.5%)	24
Sachets	3 (15%)	4 (20%)	13 (65%)	20
Tubes	1 (4.7%)	2 (9%)	19 (86.3%)	22
Wrappers	2 (15.4%)	4 (30.8%)	7 (53.8%)	13
Total	43	94	321	458

Packing Material FMS Categorization

FMS Analysis – Frequency			
	F	M	S
Corr. Boxes	> 208	207 – 101	< 100
Bags	> 427	426 – 43	< 42
Caps	> 573	572 – 100	< 99
Cartoons	> 230	229 – 103	< 102
Containers	> 220	219 – 101	< 100
Dividers	> 201	200 – 135	< 134
G/Bottles	> 200	199 – 127	< 126
I/Lining	> 161	160 – 101	< 100
Inserts	> 612	611 – 101	< 100
N/tags	> 55	54 – 11	< 10
Pouch	> 75	74 – 50	< 49
Sachets	> 75	74 – 50	< 49
Tubes	> 240	239 – 101	< 100
Wrappers	> 135	134 – 55	< 74

Table 2.6: FMS Analysis for RM & PM

FMS categorization is based on the frequency of unit transactions from September 2017 to August 2018 and obtained the results in the above two tables. The first table shows the number of SKUs and the percentage of total SKUs which belongs to each category under the stratification of FMS. The ‘F’ category represents the fast moving SKUs, while the ‘M’ category and the ‘S’ category represent the medium moving and slow moving SKUs respectively. The second table shows the SKU issued frequency for each category under FMS stratification.

2.6.2.3 Amalgamated Analysis & Inventory Layout

On their own, the ABC and FMS categorizations provide limited insight into the behavior of SKUs. Therefore, we amalgamated the matrices in order to provide better visibility into the product behavior and allocate space for each SKU. I have included my findings in the following table and this matrix amalgamates the ABC and FMS categorizations for the RM and PM. In terms of a demand planning approach, AF, AM, BF are critical categories to be considered, in order to avoid out of stocks situations.

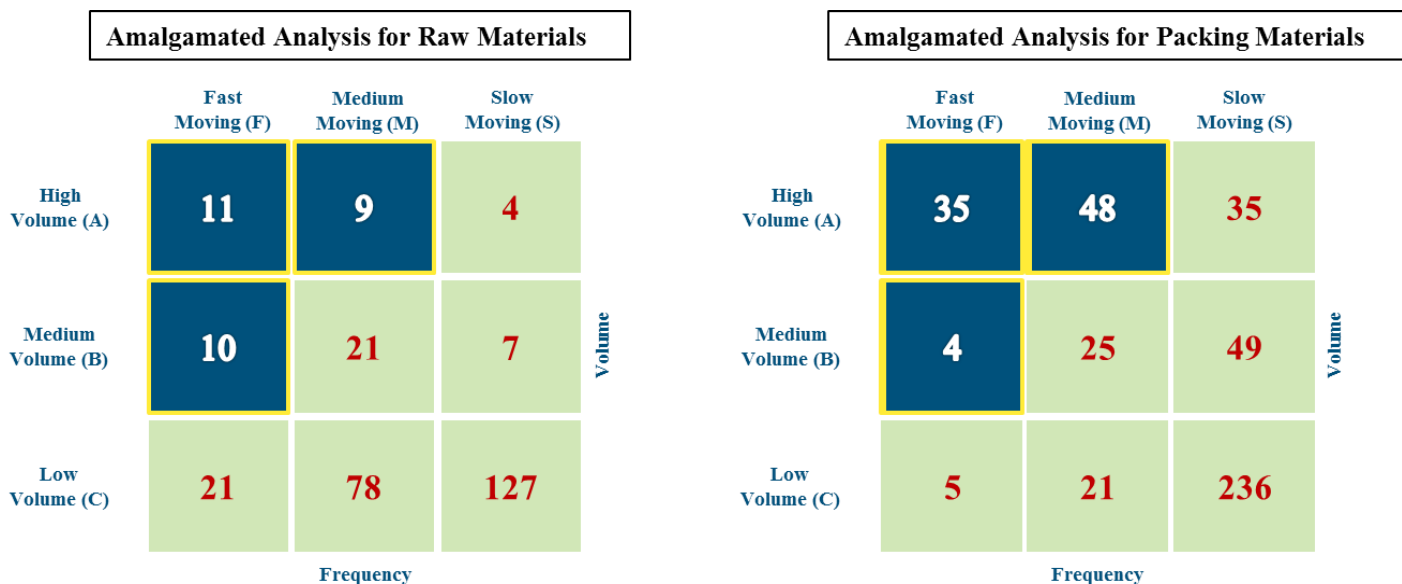


Figure 2.16: Amalgamated Analysis

As per the amalgamated analysis, I suggested following storing model to in line with “Lean” concept. That is because, according to lean low travel distance, space allocation and shorter picking time can be gained through following recommended solution.

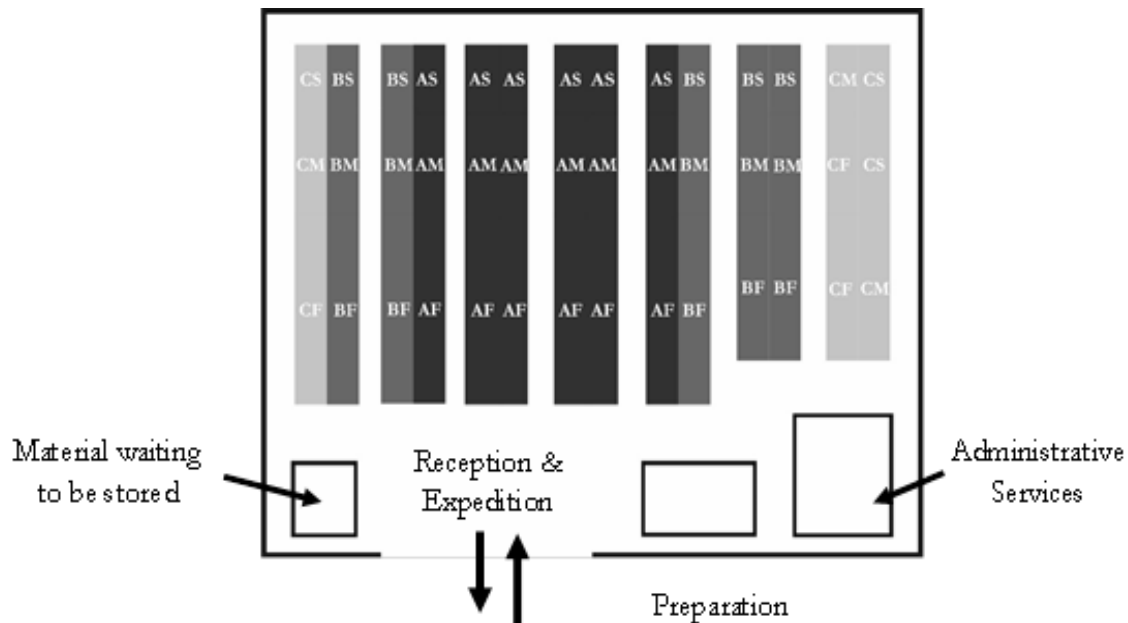


Figure 2.17: Suggested Inventory Layout

“Within-aisles storage” method is suggested as new inventory layout. According to this new layout, space allocation will be based on value and frequency of the product.

2.7 Project 3 – Inventory Optimization

2.7.1 Objective of the Project

During my internship, I was introduced to existing inventory norms and with the help management I had to change it and control the flow of materials with my fellow friend Banusha. She took care of the controlling process while I had to do the reporting part. Optimization was not happening and there were lots of drawbacks specially with higher stocks situation. As of end of first 3 months of my internship, production needs are well covered, not 100% in time, but the reports showed high availability. Therefore, the main objective of this project was **“Is it possible to reduce the surface used by safety stock level through optimization”**.

As the way forward plan, I recognized that change in the Safety lead time parameter (14 days for Packing materials and 1 month for Raw materials) can leads to inventory optimization.

2.7.2 Outcome of the Project

According to the norms and given data, I carried out following calculation and was able to add value to existing operation.

Definitions

Ideal Space Requirement = Existing norm (2 Weeks requirement for PM+ 1 Month requirement for RM)

Maximum space requirement = Buffer stock+ Order quantity

Total Warehouse capacity = Existing Total Warehouse capacity

Actual Monthly Average of Total PP = Actual monthly pallet consumption

Month	Ideal Space Requirement	Max Space Requirement	Total Warehouse Capacity	Actual Monthly Average of Total PP	Total Average deficit of Space against Actual	Total Average deficit of Space against Max Space	Gap Between Actual Deficit & Calculated Deficit
August	1,999	3,077	2,400	2586	186	677	491
September	2,026	3,158	2,400	2289	(111)	758	869
October	1,972	3,134	2,869	2041	(828)	265	1,093
November	1,840	3,092	2,869	1820	(1,049)	223	1,272

Table 2.7: Variation in Number of Total Pallet Positions

Above figure describes how total number of pallet positions varied with the impact of inventory optimization technique. In the month of October, Ideal total space requirement for both RM and PM stated as 1972 PP. But, Actual consumed PP accounted as 2041. That shows the deviation in procurement department and as the warehouse we had to keep excess stock that the ideal amount. This deviation was really high in the earlier months.

Since we are involving in the controlling process, during the November Actual Monthly Average consumption (1820 Pallet positions) was lower than the ideal space requirement (1840 Pallet Positions). That shows, how much improvement we delivered to warehouse operation using our project. Following figure describes it more comprehensively with reference to RM and PM separately.

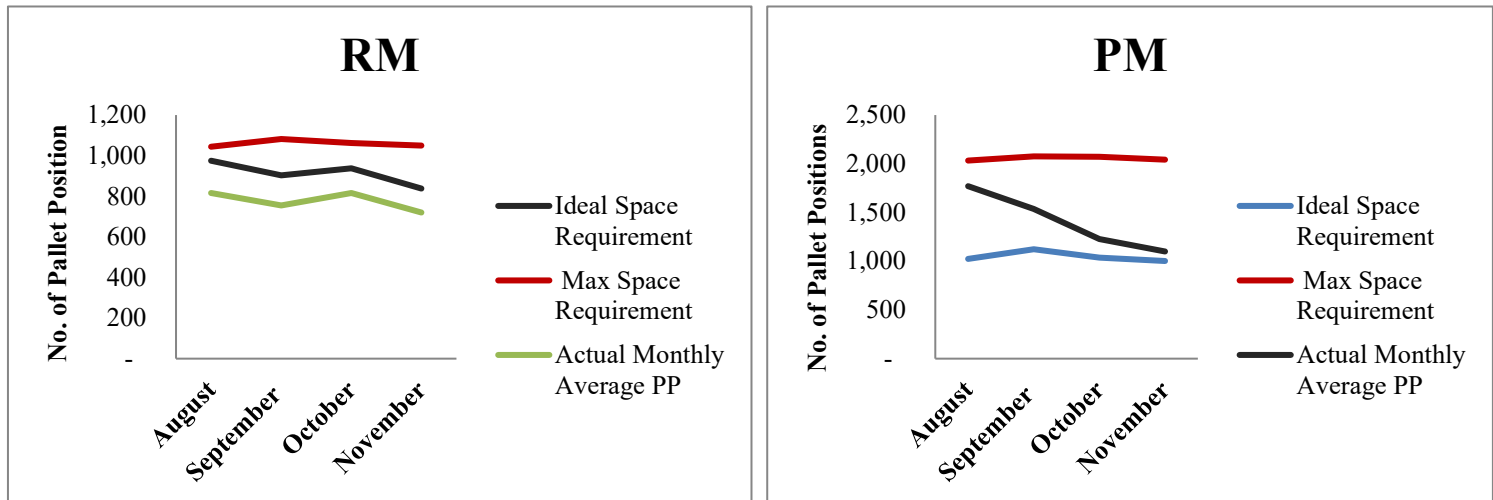


Figure 2.18: Separated Display of PP Variation

As it can be clearly seen in the above figure, Actual Monthly Average PP (Actual PP consumption) of Packing materials has touched the Ideal space requirement in the month of November. We controlled the receiving process by not accepting deliveries if we had the enough stock according to our norms. We prepared excel sheets to monitor this behavior and we did not accept any deliveries if we had enough stocks. Since most of the raw materials are imported items, we had a poor influence over RM. That is why we couldn't take that trend downside for RM. But, since more than 90% of packing materials are local items, we capitalized that opportunity and brought that trend in line with Ideal space requirement. .

2.8 Project 4 – Centralization of Labeling Process

2.8.1 Project Background

As mentioned in the figure 2.8, decentralized labeling operation at Dankotuwa and Welisara has brought in inefficiencies of low utilization of the Premier machine and increased manual labeling. I was assigned to this project at the beginning of September and I had to carry out a comprehensive analysis with top management. I was exposed to different analyzing techniques and Value Stream Mapping (VSM) is one of them.

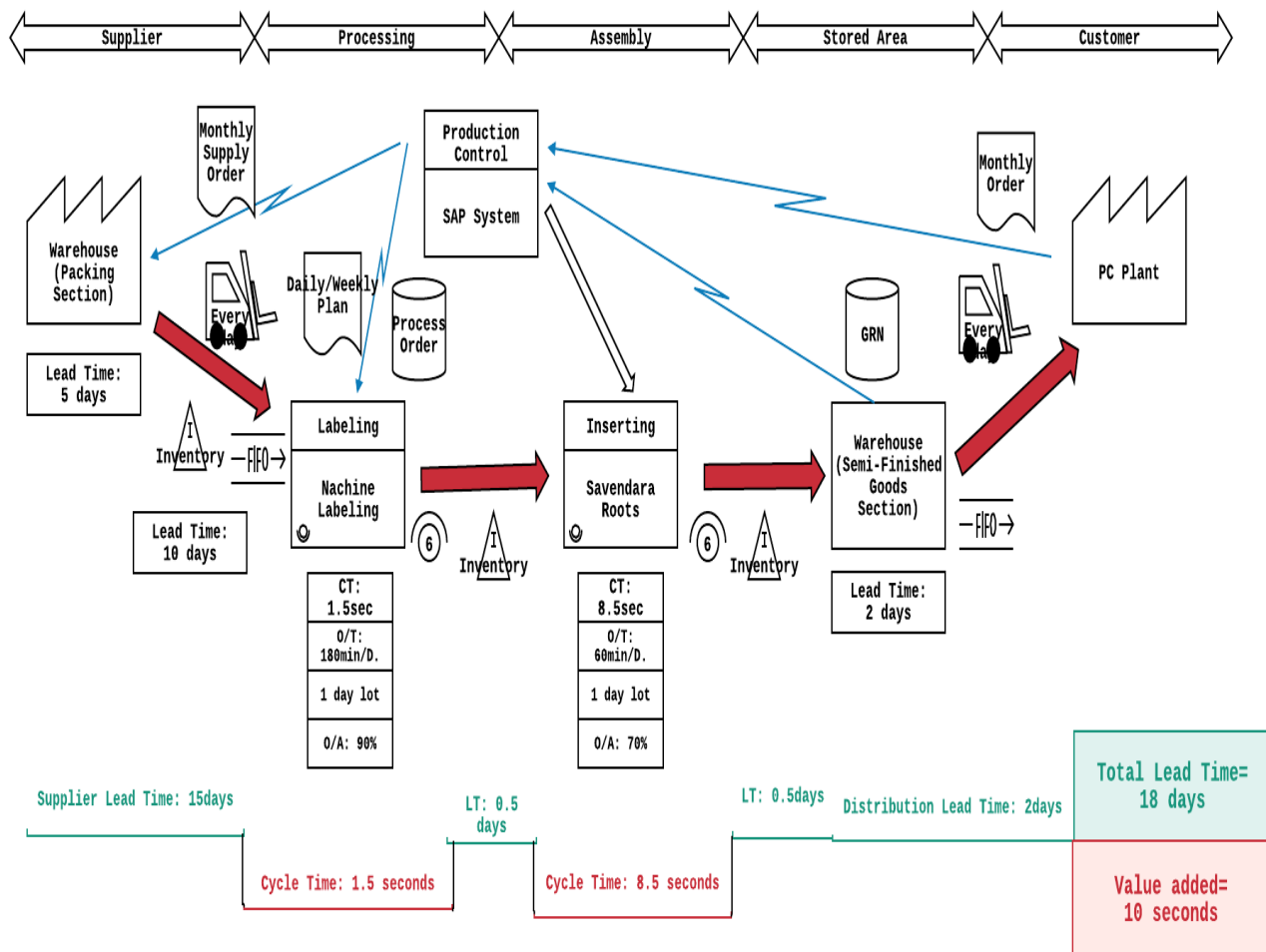


Figure 2.19: VSM for Existing Labeling Operation

As it is described in the above diagram, Value chain lasts only for 10 seconds but total lead time accounts for 18 days. With this base, I concentrated more in this issue and finally able to find the root cause using a “Cause-Effect diagram”. Main purpose of this project was to **“Centralize the labeling operation to focus more on core activity (To seek cost efficient labeling operation)”**.

Premier machine which was located inside the Personal Care plant was totally underutilized and there were more drawbacks within the existing system. As the first step I was targeting on optimizing the utilization of the Premier machine to 90% by transferring some of manually pasted labels from Welisara. Following figure describes possible root causes that restrain optimization of existing labeling process.

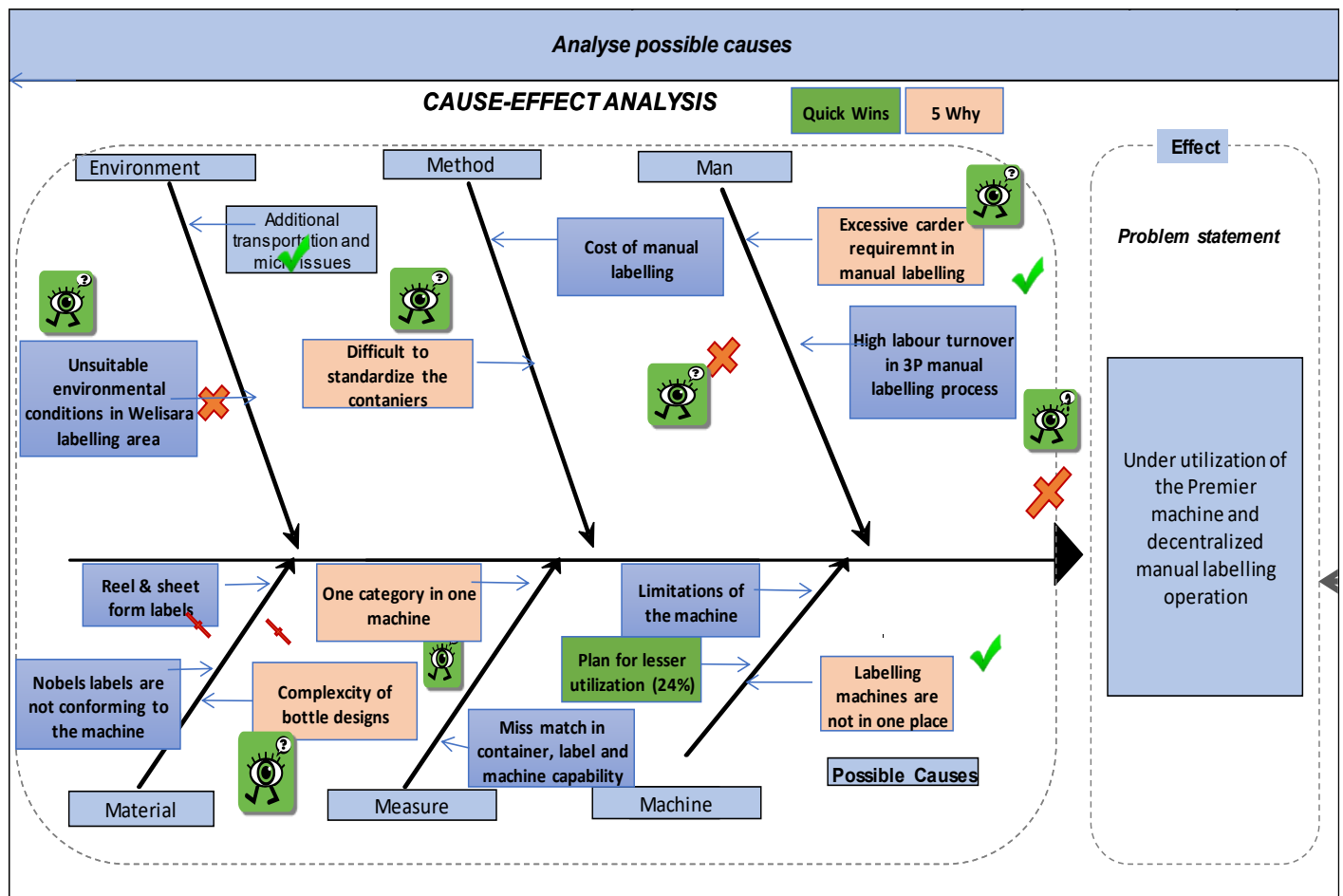


Figure 2.20: Cause-Effect Diagram

2.8.2 Improvement Opportunities

As the next step, I carried out a situational analysis using a SWOT diagram. Since there was a capability in Premier machine I conducted trials (refer appendix 01-Trial plan) to transfer SKUs from manual labeling to Machine labeling. As the sourcing strategy, I tried to check the possibility of getting labeled containers from Phoenix and Polypack suppliers. Hemas Transportation Management System (HTMS) is an advanced platform to track the shuttle cost that incurred due to labeling operations.

As per the situational analysis, I mainly focused on labeling strategy, sourcing strategy, and supply chain visibility to improve excising bottlenecks.

FOCUS AREA	ALIGNMENT OF SWOT		INITIATIVE STRATEGY
Labeling Strategy	Strength	Weakness	Conduct trials to transfer SKUs to Premier machine from Manual Labeling.
	Capability of Labeling Machine	44% Utilization & Higher Labor Cost	
Sourcing Strategy	Strength	Opportunity	Check the possibility of getting labeled Containers.
	Pool of Suppliers	In house Capability in Suppliers	
Modern Technology	Weakness	Opportunity	Use of skilled labors to operate under utilized machines.
	Operator's Lack of Knowledge	Technical Sources	
Supply Chain Visibility	Strength	Threat	Track the shuttle cost that incurred due to labeling operations
	Advanced TMS	Additional Transport Charges	

Table 2.8: SWOT Diagram

2.8.3 Outcome of the Project

As per my workings on above focused areas, following solutions were made with the help of top management.

Assumptions: An FMJ supplier does not have an existing labeling machine. Therefore, labeled containers can be negotiated only with Phoenix and Polypack suppliers. 15 SKUs are identified as export Kumarika SKUs which are going to be transferred to Top Print labeling.

As per the supplier agreement, Baby Cheramy Cream range items can be outsourced to Phoniex supplier. They would send labeled containers and Top Print machine will be shifted to FMJ suppliers with the total Kumarika range. It will provide a competitive advantage to Hemas because all the Kumarika containers are manufactured by FMJ supplier. Therefore no additional cost is included in terms number of trips and Hemas can focus more on their core business instead of spending time on doing labeling process. New utilization levels are also included separately for each machine.

	Current SKUs	OutPut	Utilisation	Proposed SKUs	Out Put	utilisation	Moving into manual from current list - Skus	Moving into phoenix - SKUs	Moving into Poly packs - SKUs
Komark	53	1,225,360	98% (In 24hours shift)	7	541,799	95% (In 12hours shift)		13	
Premier	15	431,760	23% (In 12 hours shift)	39	480,667	85% (In 12hours shift)		9	
Top print	6	477,501	83% (In 12 hours shift)	21	899,191	156% (In 12hours shift)/ 74% (In 24hours shift)			

Table 2.9: Summarization of New Composition

2.9 General Works

The four main projects I involved in was spanning throughout my 24 weeks at Hemas. In addition to them, I was involved in few routing activities to support the operations at the warehouse. These additional work tasks gave me a wider exposure to the corporate work world and also to extensive Warehouse Operational practices at Hemas.

2.9.1 Over-time Calculation

I was assigned to derive an overtime norm in order to capitalize warehouse costing operations. Warehouse management is following a cost cutting method and I was asked to calculate single Overtime hours and Double Overtime hours in the existing carder to monitor human allocation for each department. There, I got an opportunity to work Human Resource department and about payment methods. In addition, I was asked to prepare a presentation by taking confidential information using the SAP system. When doing the overtime calculation, I followed these assumptions with the guidance of top management. Following graph describes how Single and Double OT varies according to each department.

Assumptions

1. One month carder is different from the other.
2. In a particular month, there are 20 weekdays+ 4 Saturdays+ 1 Sunday is allowed to work
3. Casual workers are paid double hours only on Poya days.

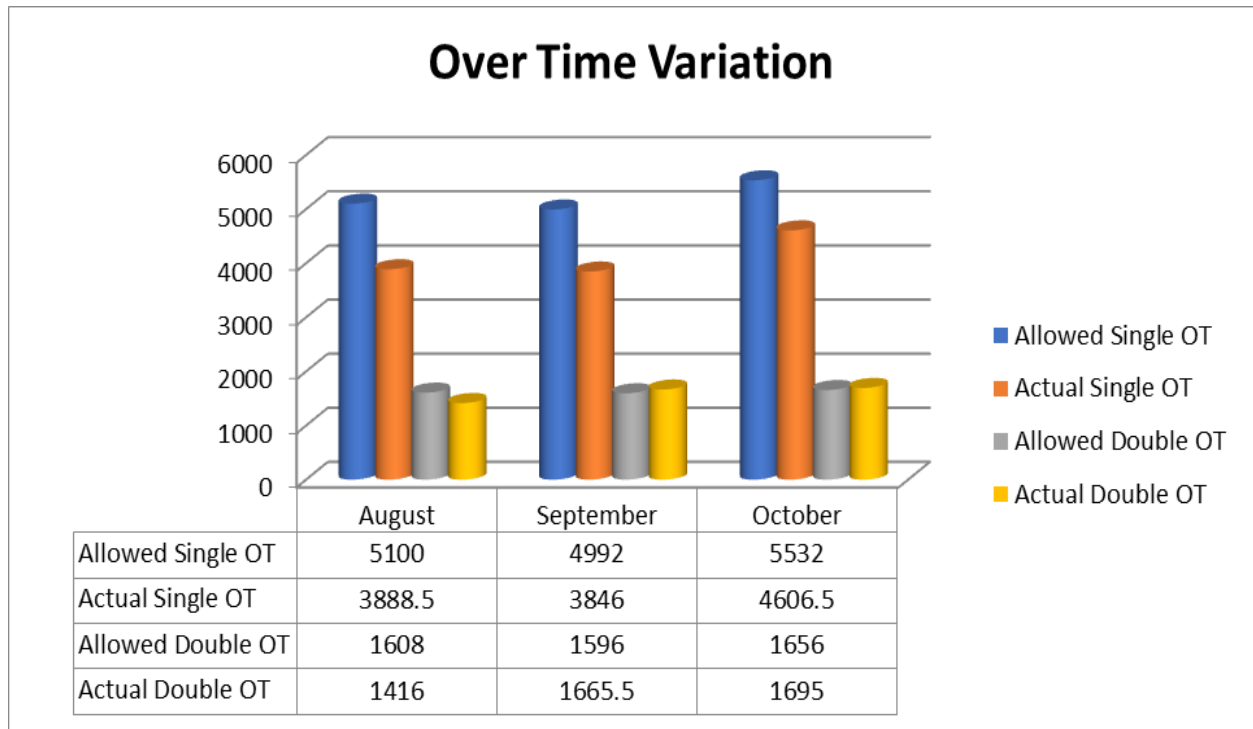


Figure 2.21: Comprehensive Overtime Calculation

2.9.2 Plant Visits and Audits

Throughout the internship I was able to visit the whole Hemas manufacturing complex. Mostly I visited the Personal Care plant and Procurement division. During the past visits, I was able to get exposure to manufacturing operations and specially the change overs time etc. One of the most important impact of these visits is that I was able to engage with people specially with their day to activities.

- 1. Engagement through Activities-** Activities included within the calendar year.
- 2. Engagement through Talent Development-** Under the Talent Society (Bera, Drama, Musical Band & Dancing Team.
- 3. Engagement through Appreciation-** Star of the month programme to identify the best Kizen suggestion (No of Total Kaizens in 2017/18 - 64), Birthday Celebrations at Functional Meetings.
- 4. Engagement through Employees Forum-** Employee Advisory Committee Meeting, Transport Committee Meeting, Facilities Committee Meeting, Morning Stand Up Meetings, Monthly Functional Meetings etc.

Plant	PC	Soap	SanNap	Diva (3P)
				
Product Categories	Oral care, Skin care, Hair care, Baby care, Cologne & Perfumes	Baby & Beauty Soap	Sanitary Napkins	Detergent powder, Laundry soap, Baby Diapers, Talc, Nappy wash
SKUs	192	19	8	24
HC	195	77	40	4

Figure 2.22: Dankotuwa Manufacturing Complex

In addition to those engagements, I was able to participate with several internal and external audits in different time periods. Hemas is mainly concerned about their health and safety procedures they have complied with **OHSAS 18001 – Occupational Health & Safety Assessment Series** (Assuring Health & Safety of all the stakeholders of the Organization), **ISO 14001 – EMS** (Minimizing the environmental impact due to company operations and prevention of environmental pollution) and **ISO 9001-2015** (Certificate for the Good Manufacturing Practices Systems). There were different types of audits specially GMP (Good Manufacturing Practice) and GWP (Good Warehousing Practice) audits.

CHAPTER 03- CONCLUSION

3.1 Training Experience

As a Transport and Logistics Management undergraduate it is very important to gain knowledge and experience in various aspects from industry. Therefore the industrial training really added value since it gave a substantial opportunity to practice and experience real supply chain and logistics applications as well as management practices in the FMCG industry which is a totally different experience compared university. I consider the internship at Hemas manufacturing plant is a life changing opportunity where I could learn as well as apply my theoretical knowledge in the Sri Lanka's largest chain of FMCG products (As a local organization). As Hemas FMCG sector has a differentiate product portfolio of 15 brands, it is like enormous number of supply chains under one umbrella which is very treasured experience for a transport and logistics industrial trainee.

During "Centralization of Labeling Project" I got an opportunity to work with top management and got an exposure to attend those management meetings. With this project, I was able to identify how the efficiency of the processes can be enhanced through proper planning and root cause analysis. In addition, I got to know the practical issues in implementing such project and constraints in outsourcing an existing project, specially with reference to engineering and quality department. On the other hand, I had to coordinate with different parties to conduct labeling trails and that provided me a great opportunity to understand the core business and key requirements I should have. Sometimes I had to send Reservation Slips to Welisara plant asking different labeling containers. Working with different kind of people and introduction to SAP system were great opportunities I got through my internship program.

In addition to those inputs, training period also gave me an opportunity to learn Excel Applications in Analytics in a more comprehensive way. While analyzing the findings of the inventory reports I was able to explore several areas in Excel which I did not have knowledge before. When I was calculating unit cost, I got an exposure with different cost calculations methods and in house project reports types such as 'A3 report format'. Exposure to work as an employee was another opportunity I got because I was able to participate in different audits was able to familiarize with audit procedures.

3.2 Training provided by Hemas

According to my perspective, I think Hemas is one of the best organizations where an industrial trainee can get a good industrial experience as well as the exposure to local organizational practices. Any transport and logistics management undergraduate can get a great exposure on the Hemas logistics operations specially within the warehouse. There were 10 undergraduates from engineering faculty and we all had given projects to complete before the end of internship. Specialty in Hemas management is that they provided a great opportunity to present our project results to the Hemas Leadership Team (Board of LT) that includes Managing Director, 6 Directors, Senior Managers and Managers.

Uniqueness of Hemas was that they invited our department lecturers to participate and witness our skills in that final presentation ceremony. Head of Transport and Logistics Department Prof. Amal S Kumara and Senior Lecturer Doc. Ranil Sugathadasa participated for this final presentation program and that was really passionate about Hemas management. In addition to all the opportunities, they provided a really good platform to express ourselves to department and it allowed creating a really good bond between Hemas management and our Transport and Logistics Management department. I think this is a unique experience and a great advantage for an industrial trainee who trains at Hemas.

3.3 Suggestions for University and NAITA

The Industrial training program which is organized by NAITA is a great opportunity for third year undergraduates, of University of Moratuwa to get the exposure to their related industries and improve the practical experiences in the engineering perspective. I could explore the color of logistics industry through the internship in my third year which motivated me to specialize in this field in my final year. Even this exposure improved my curiosity to study and research more in the field of FMCG logistics. I really appreciate the NAITA's contribution towards the young professionals of the country, because they can show the hidden opportunities in different fields before completing their degree programs.

Visits of officials during our training period are something which is beneficial for both the trainee and the University and NAITA for the assessment process. Since Hemas manufacturing plant is located in Dankotuwa area, travel distance sometime could become a barrier as well. We had one visit from NAITA and it was a really good motive to understand the working environment. We were informed earlier by NAITA and that provided easiness when arranging the meeting and allocating the time.